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Reclamation of tidal flats to paddy soils reshuffles the soil microbiomes along a 53-year reclamation chronosequence: Evidence from assembly processes, co-occurrence patterns and multifunctionality

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ABSTRACT

Coastal soil microbiomes play a key role in coastal ecosystem functioning and are intensely threatened by land reclamation. However, the impacts of coastal reclamation on soil microbial communities, particularly on their assembly processes, co-occurrence patterns, and the multiple soil functions they support, remain poorly understood. This impedes our capability to comprehensively evaluate the impacts of coastal reclamation on soil microbiomes and to restore coastal ecosystem functions degraded by reclamation. Here, we investigated the temporal dynamics of bacterial and fungal communities, community assembly processes, co-occurrence patterns, and ecosystem multifunctionality along a 53-year chronosequence of paddy soil following reclamation from tidal flats. Reclamation of tidal flats to paddy soils resulted in decreased β-diversity, increased homogeneous selection, and decreased network complexity and robustness of both bacterial and fungal communities, but caused contrasting α-diversity response patterns of them. Reclamation of tidal flats to paddy soils also decreased the multifunctionality of coastal ecosystems, which was largely associated with the fungal network complexity and α-diversity. Collectively, this work demonstrates that coastal reclamation strongly reshaped the soil microbiomes at the level of assembly mechanisms, interaction patterns, and functionality level, and highlights that soil fungal community complexity should be considered as a key factor in restoring coastal ecosystem functions deteriorated by land reclamation.

1. Introduction

Coastal wetlands serve as key transitional zones from terrestrial to marine ecosystems and provide exceptionally valuable ecosystem services (Wang et al., 2023; Wang et al., 2021b). However, coastal wetlands are under threat from both climate change (e.g., sea level rise and storms) and anthropogenic activities (e.g., land reclamation, eutrophication and species invasion) (Fluet-Chouinard et al., 2023; Murray et al., 2019; Blum and Roberts, 2009). Among these, land reclamation is considered as one of the most important drivers of the rapid degradation of coastal ecosystems (Wang et al., 2023; Wang et al., 2021b), which results in reduced biodiversity, degraded water quality, and weakened carbon sink capacity (Wang et al., 2023; Tan et al., 2020; Yang et al., 2022b). Given its impact on sustainability of coastal ecosystems, several countries have implemented policies and regulations to manage reclamation projects and control the scale of reclamation, such as the "National Marine Function Zoning Plan" in China, the "Integrated Coastal Zone Management" strategy in Europe, and the "no-net-loss" policy in the United States (Wang et al., 2023; Ma et al., 2014).

Soils microbiomes are highly diverse and complex (Banerjee and van der Heijden, 2023), which comprise up to one quarter of Earth's diversity (Wagg et al., 2019). This vast diversity plays a key role in sustaining multiple ecosystem functions and services simultaneously (i.e., multifunctionality) by supporting multiple ecological processes, such as

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primary production, organic matter decomposition, nutrient cycles, and climate regulation (Kong et al., 2022; Jiao et al., 2022; Hu et al., 2021). The diversity, composition, and functionality of soil microbiomes are sensitive to anthropogenic land-use changes (Zhou et al., 2020; Kong et al., 2022). This has also been verified in coastal ecosystems. Previous studies have shown that reclamation of coastal wetlands to agricultural land uses strongly changed the diversity and structure of microbiomes (Ding et al., 2017; Cui et al.,2012; Wang et al., 2021a) and associated ecological functions (Jiang et al., 2021; Tan et al., 2022; Tan et al., 2020; Yang et al., 2022b). However, most of these previous studies failed to provide a mechanistic understanding of how coastal reclamation influences soil microbiomes because they did not uncover the ecological processes that underpin microbial community variations. Two types of processes, i.e., deterministic (niche-based) and stochastic (neutral) processes govern microbial community variations (Ning et al., 2019; Stegen et al., 2013). This lack of mechanistic understanding impairs our capability to determine the linkages between microbiomes and ecosystem functions after coastal reclamation (Zhou et al., 2017; Liu et al., 2021).

Apart from high diversity, another feature of the soil microbiomes is their diverse ecological relationships, ranging from mutualism to predation and competition (Zhou et al., 2021; Yuan et al., 2021; de Vries et al., 2018). Such complicated interactions can be represented as co-occurrence networks with microbial taxa as nodes and their relationships as links (Yuan et al., 2021; Zhou et al., 2021; Liu et al., 2020a). Although the co-occurrence network analysis does not necessarily reflect true interactions (Qiu et al., 2021; Liu et al., 2020a), it helps to understand the complexity and robustness of microbiomes and how these properties may vary in response to environmental change and how microbe-microbe associations may have implications for ecosystem functioning (Chen et al., 2022e; Morrien et al., 2017; Qiu et al., 2021; Wagg et al., 2019). For instance, global changes, including warming, drought and land-use change, were confirmed to change the network structure and robustness of soil microbiomes, with consequences for ecological functions, such as soil carbon cycle and multifunctionality (Yuan et al., 2021; de Vries et al., 2018; Wu et al., 2021; Kong et al., 2022). However, it is not clear whether and how coastal reclamation influences network complexity and robustness of soil microbiomes, and how this relates to soil multifunctionality in coastal ecosystems. In order to restore the coastal ecosystem functions degraded by land reclamation, it is essential to gain a comprehensive understanding of the changes in associations and functions among microbes induced by land reclamation.

To address the above knowledge gaps, this work examined the temporal dynamics of diversity, composition, ecological assembly processes and network structure of bacterial and fungal communities along a reclamation chronosequence in coastal ecosystems. The chronosequence soils spanned reclamation periods of approximately 0, 6, 23, 33, and 53 years. We also tested whether reclamation-induced changes in microbial communities are associated with the multifunctionality of coastal ecosystems. Reclamation of coastal wetlands to paddy soils was taken as an example because it is a long-term and prevalent agricultural practice in coastal reclamation, especially in many coastal regions of China (Ding et al., 2017; Jiang et al., 2021). The specific aims of this work were to (i) identify the ecological assembly processes that drive microbial community succession along a reclamation chronosequence in coastal ecosystems; (ii) investigate the temporal dynamics of microbial network structure along a reclamation chronosequence; and (iii) explore whether and how changes in microbial communities following land reclamation influence the multifunctionality of coastal ecosystems. This work provides a comprehensive insight into how land reclamation influences the soil microbiomes in coastal ecosystems, and has broader value for understanding and predicting how coastal ecosystem microbiomes and functions respond to land reclamation at globally relevant scales.

2. Materials and methods

2.1. Study area and field sampling

This study was conducted in Chongming Island (121.87°E − 121.98° E, 31.48° N −31.52° N; Fig. S1 and Table S1), Shanghai, China. Located at the Yangtze Estuary, Chongming Island is the largest coastal island in China and approximately half of its present area was reclaimed from coastal wetlands (Jiang et al., 2021). The study region is characterized by a semitropical monsoon climate. The mean temperature and mean annual precipitation are 16.0 ℃ and 1144 mm, respectively. The texture of the tidal flat is mud and fine sand, while the texture of paddy soils reclaimed from tidal flat is sandy loam.

Soil samples were collected from a land reclamation region in Chongming Island in July and December 2021. A methodology of space-for-time analysis was used to identify soil reclamation chronosequence (Ding et al., 2017; Wu et al., 2021). The sampling region spanning < 12 km include tidal flats (DT), reclaimed tidal flats for 6 years but not cultivated (ST0), and paddy soils that have been reclaimed and cultivated for 23 (ST1), 33 (ST2), and 53 years (ST3) (Fig. S1) (Jiang et al., 2021). Within each sample area, 4 sub-sites were selected to collect soil samples from six separate cores (0–10 cm) at each sub-site and mixed to constitute one composite sample per sub-site. In total, 40 (5 sample areas × 4 sub-sites × 2 time points) composite soil samples were taken and transported to the laboratory within 3 h. The collected soil samples were subsequently homogenized and divided into two subsamples. One part of subsample was stored at −20 °C for measurement of soil physiochemical parameters and DNA extraction, and the remaining part was preserved at 4 °C for determination of extracellular enzyme activities and potential microbial decomposition.

2.2. Measurement of physicochemical parameters, extracellular enzyme activities, and potential microbial decomposition

Soil pH was determined with a pH meter (Mettler-Toledo, Switzerland) in a soil: water extract (1:2.5) (Chen et al., 2022a). Electrical conductivity (EC) was detected through a 2265FS EC meter (Spectrum Technologies Inc., USA) (Gao et al., 2017). Moisture (MS) was measured using oven-drying soils (105 °C) method (Kong et al., 2022). Ferrous oxides (Fe^{2+}) and ferric iron (Fe^{3+}) were measured by colorimetric method (Haese et al., 1997). Soil total organic carbon (TOC) and nitrogen (TN) were measured on a TOC automatic analyzer with an SSM-5000A detector (Shimadzu, Japan) (Chen et al., 2022b). Soil nitrate (NO_3^--N), nitrite (NO_2^--N), and ammonium (NH_4^+-N) in the KCl extracts were measured by a continuous analyzer (SKALAR SAN ++, Netherlands) (Gao et al., 2020). Soil total phosphorus (TP) and available phosphorus (AP) were measured by molybdenum blue method after perchloric acid digestion and $NaHCO_3$ extraction, respectively (Sommers and Nelson, 1972; Hu et al., 2021). Soil total sulfur (TS) was detected by alkaline digestion, as described by Churka Blum et al. (2013). Available sulfur (AS) was measured by turbidimetry after ammonium acetate extraction (Wang et al., 2016).

Six extracellular enzyme activities involved in C cycle (β-1,4-glucosidase [BG] and β-D-cellobiohydrolase [CBH]), N cycle (β-1,4-N-acetylglucosaminidase [NAG] and L-leucine aminopeptidase [LAP]), P cycle (alkaline phosphatase [APO]), and S cycle (arylsulfatase [ASF]) were detected using 96-well microplates according to the method previously described (Kong et al., 2022; Kunito et al., 2022). Potential microbial decompositions (organic carbon and nitrogen mineralization) were determined by incubating 30 g of fresh soil samples at 25 °C in 50 ml plastic specimen cups for two weeks in the dark (Page et al., 1982; Han et al., 2022b). Cumulative organic carbon mineralization (Cm) was calculated by summing the total amount of CO_2 released from soils during the incubation period (Kong et al., 2022; Fanin et al., 2018). Net N mineralization (Nm) was estimated as the difference between initial and final inorganic N (sum of NH_4^+-N and NO_3^--N) before and after

incubation (Kong et al., 2022; Jiao et al., 2022).

2.3. DNA extraction, high-throughput qPCR, and amplicon sequencing

Total genomic DNA of each sample was extracted from 0.25 g of soils using the DNeasy PowerSoil Kit (QIAGEN, USA) according to the manufacturer's manual. DNA quantity was assessed using a NanoDrop-2000 spectrophotometer (Thermo Scientific, USA). High-quality DNA was used to quantify functional genes related to C, N, S, P cycles and one 16S rRNA gene using Wafergen SmartChip Real-time PCR System. Detailed information about primers, the specific reaction system and amplification conditions were described by Wang et al. (2022). The high-throughput quantitative PCR (qPCR) results were analyzed using SmartChip qPCR Software (v 2.7.0.1). The amplification efficiencies were set in the range of 0.90–1.1. The relative abundance of each functional gene was calculated via dividing the functional gene copy number by the 16S rRNA gene copy number (Su et al., 2021a). Finally, 25 functional genes were used in this work (Table S2) because they were detected in all samples and represented a high quality of data.

High-throughput sequencing was performed on the Illumina MiSeq PE300 platform. The primer 338F (5′-ACTCCTACGGGAGGCAGCA-3′)/806R (5′-GGACTACHVGGGTWTCTAAT-3′) was applied to amplify the V3–V4 regions of the bacterial 16S rRNA gene (Jiang et al., 2022), and ITS1F (5′-CTTGGTCATTTAGAGGAAGTAA-3′)/ITS2R (5′-GCTGCGTTC TTCATCGATGC-3′) was applied to amplify the fungal ITS1 gene (Luo et al., 2023). The 16S rRNA and fungal ITS1 gene sequences were processed with QIIME (v.1.9.1) and USEARCH (v.11.0). Paired-end sequences were merged using Flash (v.1.2.11) and then quality-filtered based on the procedure elaborated by Caporaso et al. (2011). Chimera detection and removal were performed using the USEARCH algorithm (Edgar et al., 2011). The UPARSE pipeline (v.11.0) was applied for operational taxonomic unit (OTU) clustering with a sequence at a 97% similarity level (Edgar, 2013). OTUs lacking more than two sequences were removed; for each representative sequence, the SILVA (release 138; for 16S) (Quast et al., 2013) and UNITE (v.8.0; for ITS) (Nilsson et al., 2019) databases were applied to assign the taxonomic annotation using the RDP classifier. The raw sequence data are available in the NCBI Sequence Read Archive (SRA) database (PRJNA868926 for 16S rRNA sequences and PRJNA868931 for ITS sequences).

2.4. Ecosystem multifunctionality assessment

Ecosystem multifunctionality (EMF) was assessed using the averaging approach, which has been widely applied in the multifunctionality literature (Hu et al., 2021; Qiu et al., 2021; Jiao et al., 2022; Fan et al., 2018). Twenty of the soil indictors detected were selected to assess the EMF. The soil metrics included soil TOC, BG, CBH, Cm, TN, NH_4^+, NO_3^-, NAG, LAP, Nm, C/N, TP, AP, APO, TS, AS, ASF, Fe^{2+}, Fe^{3+}, and 25 functional genes related to C, N, S, and P cycles. These indicators cover a range of soil processes, including organic matter decomposition, nutrient pool dynamics, and biogeochemical cycles, which were regarded as critical determinants of ecological functioning in coastal ecosystems (Li et al., 2022a; Sun et al., 2023; Zhang et al., 2021b). The 20 indicators were first standardized individually ranging from 0 to 1 and then averaged to obtain an EMF index.

2.5. Data analysis

Data analysis in this work was conducted in the R environment (v.4.1.2) using the packages "vegan", "ggplot2", "reshape2", "stats", "pheatmap", "igraph", "dplyr", "rdacca.hp", "spaa", "iCAMP", "WGCNA", "rfPermute", "A3", "adespatial" and "piecewiseSEM" unless otherwise indicated.

Microbial alpha (α)-diversity including the OTU richness (richness) and Shannon index were calculated using the "vegan" package (Huang et al., 2021). We created one index to represent the microbial α-diversity (α-diversity index) through averaging the standardized scores (a common scale ranging from 0 to 1) of richness and Shannon index. Random forest models were performed to identify the main environmental predictors of microbial α-diversity. The percentage increases in the MSE (mean squared error) were used to evaluate the importance of each predictor, with higher MSE% values indicating more vital factors (Jiao et al., 2020). The significance of model and each predictor was evaluated using the "rfPermute" and "A3" package, respectively (Jiao et al., 2020). Principal coordinate analysis (PCoA) based on Bray-Curtis distance and multivariate permutational ANOVA (PERMANOVA) were employed to evaluate the dissimilarities in the microbial communities using the "vegan" package. β-diversity (Bray-Curtis dissimilarity) was then partitioned into replacement and richness difference components using the beta.div.comp function of the "adespatial" package (Legendre, 2014), and its correlations to physicochemical parameters were assessed using the Mantel test. Furthermore, species contribution to β-diversity (SCBD) was estimated using the beta.div function in the "adespatial" package (Santos et al., 2021). Distance-based redundancy analysis (dbRDA) was utilized to evaluate the relative importance of physicochemical parameters on microbial community composition (Huang et al., 2022). The contribution and significance of individual parameter to community composition was assessed by performing the rdacca.hp and permu.hp function in the "rdacca.hp" package (Lai et al., 2022). The associations between the relative abundance of microbial taxa (phylum and family level) and physicochemical parameters were also assessed by random forest models (Jiao and Lu, 2020), with physicochemical parameters serving as predicators for the relative abundance of microbial taxa.

To estimate species co-occurrence patterns along the reclamation chronosequence, Pearson correlations based on networks were constructed. The OTUs presented in more than four samples were selected to calculate the correlations. Bacterial and fungal community co-occurrence networks were created using robust ($|r| > 0.8$) and remarkable (Benjamini-Hochberg corrected $p < 0.01$) Pearson correlations, which were then visualized in the R environment. Network topological parameters, including node number, link number, degree, eigenvector centrality, and linkage density, were calculated by the "igraph" package. The topological properties of each sample (sub-network) were obtained via the subgraph function in the "igraph" package (Ma et al., 2016). The degree, eigenvector centrality and linkage density were used to reflect the network complexity (Li et al., 2022b; Chen et al., 2022d; Wu et al., 2021). We then utilized the three topological parameters from sub-network to create one index (network complexity index) to assess the complexity of networks using the method described as above (Jiao et al., 2022). Robustness and vulnerability indices were calculated to characterize the robustness of microbial networks (Chen et al., 2022d; Yuan et al., 2021). Microbial networks with high robustness but low vulnerability are considered more stable (Chen et al., 2022d; Yuan et al., 2021).

To quantify the relative importance of deterministic processes and stochastic processes in structuring bacterial and fungal communities, the Infer Community Assembly Mechanisms by Phylogenetic-bin-based null model (iCAMP) was applied. iCAMP can infer the relative contributions of five ecological processes (i.e., heterogeneous selection, homogeneous selection, dispersal limitation, homogenizing dispersal, and drift) governing community assembly, based on the within-bin beta net relatedness index and the modified Raup-Crick metric value (Ning et al., 2020). In contrast to the entire community-based approach, iCAMP quantifies the relative contributions of different ecological processes for each phylogenetic group (bin), providing a significant advance in quantitative performance (Aslani et al., 2022; Ju et al., 2021; Ning et al., 2020; Sun et al., 2021; Yang et al., 2022a). Random forest models were applied to assess the associations between the relative contribution of ecological assembly processes and the variations in physicochemical parameters. In addition, Levins' niche breadth of microbial community was calculated using the "spaa" package to evaluate habitat specialization and

elucidate the contribution of assembly processes (Zhao et al., 2022a).

To evaluate the direct and indirect impacts of reclamation-induced variations in soil physicochemical properties (EC, pH and MS), microbial diversity and network complexity on EMF, a piecewise structural equation modeling (piecewiseSEM) was established using the "piecewiseSEM" package. We developed a priori model based on assumptions and theoretical relationships between soil physicochemical properties, microbial diversity, network complexity and EMF (Fig. S2). Shipley's d-separation test was then performed to test the fit of the model, and Fisher's C statistic with $p > 0.05$ suggested a good fit (Shipley, 2009). Prior to constructing piecewiseSEM, all explanatory variables were standardized (a common scale ranging from 0 to 1) to obtain standardized path coefficients.

3. Results

3.1. Soil physicochemical parameters and multiple functions along the reclamation chronosequence

Reclamation of tidal flats to paddy soils resulted in a decline in soil salinity (indicated by EC), pH (except that in ST0), MS, and iron ions (Fig. S3). However, only EC and Fe^{2+} in all reclaimed soils were significantly different from those in tidal flats (Fig. S3). Soil carbon (TOC) and nitrogen (NH_4^+-N and NO_3^--N) components were lower in ST0 and ST1 than in DT, while they were significantly higher in ST2 and ST3 than in DT. Tidal flats reclamation increased TN and NO_2^--N, while decreased C/N ratio. The remarkable difference from DT was observed in ST2 and ST3. Soil phosphorus (TP and AP) and sulfur (TS and AS) components were overall lower in reclaimed soils than in tidal flats, except for TS in ST3, TP in ST0, and AP in ST1.

Tidal flats reclamation decreased the activities of CBH (except that in ST0), NAG, LAP, APO, and ASF but increased the activities of BG (Fig. S3). Organic carbon (Cm) and nitrogen mineralization (Nm) were significantly lower in ST0 than in DT. However, they gradually increased from ST1 to ST3, and Cm was even significantly higher in ST3 than in DT. The relative abundance of functional genes related to C fixation was slightly lower in ST0 and ST1 than in DT, while that in ST2 and ST3 was significantly higher than in DT. The relative abundance of functional genes related to C degradation was generally decreased after tidal flats reclamation except that in ST3. Furthermore, reclamation of tidal flats to paddy soils generally decreased the relative abundance of functional genes related to N (except that in ST3), S, and P cycles.

3.2. Microbial diversity and composition along the reclamation chronosequence

Bacterial α-diversity showed a slight decrease in reclaimed tidal flats without cultivation ($p > 0.05$), but increased to higher levels in paddy soils (Fig. 1a and S4). Both fungal richness and Shannon index were lower in reclaimed soils, although they showed different trends along

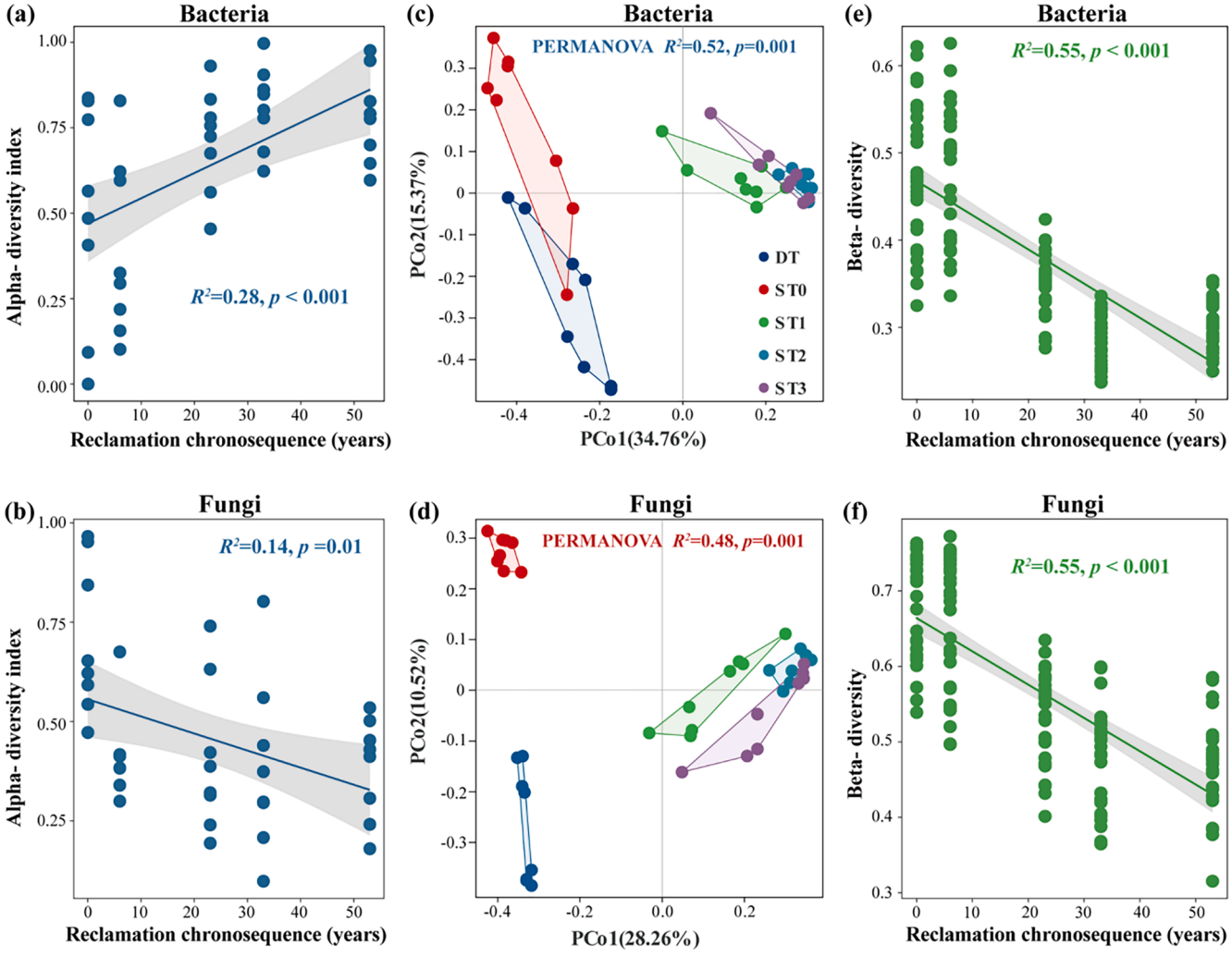


Fig. 1. Microbial diversity along the reclamation chronosequence. Bacterial (a) and Fungal (b) α-diversity index. Principal coordinates analysis (PCoA) plot of bacterial (c) and fungal (d) community based on Bray-Curtis distance. Bacterial (E) and Fungal (F) β-diversity.

the reclamation chronosequence (Fig. S4). Standardized α-diversity index further suggested that fungal α-diversity significantly decreased along the reclamation chronosequence (Fig. 1b). Random forest analysis showed that pH, EC, and NO_x^- were the main predicators of bacterial α-diversity, while EC and Fe^{2+} were relatively more important in predicating fungal α-diversity (Fig. S5). Principal coordinate analysis suggested that the compositional dissimilarities of bacterial and fungal community among reclamation chronosequence were significant (Fig. 1c and d), and microbial communities between tidal flats and reclaimed soils were distinctly separated (Table S3).

β-diversity index of bacterial and fungal community remarkably decreased along the reclamation chronosequence (Fig. 1e and f). The β-diversity decomposition analysis suggested that the compositional dissimilarities of microbial communities were dominated by species replacement processes throughout the reclamation chronosequence (contributing 71.2% and 75.29% to bacterial and fungal β-diversity, respectively), while richness difference processes contributed only 28.8% and 24.7% on average (Fig. S6a and b). The Mantel test showed that bacterial and fungal β-diversity were mainly influenced by the changes in EC, pH, and AP (Table S4).

3.3. Microbial taxa abundance and associated key drivers along the reclamation chronosequence

Bacterial community composition primarily comprised of *Actinobacteriota*, *Firmicutes*, *Proteobacteria*, *Chloroflexi*, and *Acidobacteriota* phyla (Fig. 2a). With the exception of *unclassified_k_Fungi*, *Ascomycota*, *Chytridiomycota*, *and Basidiomycota* were the dominant phyla in the fungal community (Fig. 2b). While the dominant phyla of bacterial and fungal communities were similar throughout the reclamation chronosequence, the relative abundance of some dominant phyla varied remarkably along the reclamation chronosequence. In particular, the relative abundance of *Firmicutes* remarkably decreased after tidal flats reclamation except for a slight increase in ST0 ($p > 0.05$). The relative abundance of the dominant fungal phylum, *Ascomycota,* markedly increased after tidal flats reclamation, while the relative abundance of *Chytridiomycota* significantly decreased.

Results from RDA, variation partition and random forest analysis showed that EC and pH were the relatively more important environmental factors for bacterial and fungal community variations at OTU, phylum, and family level (Fig. 2c, 3 and S8). Therefore, reclamation of tidal flats to paddy soils resulting in shifts in microbial community structure, was remarkably associated with the changes in EC and pH.

3.4. Ecological assembly processes along the reclamation chronosequence

Stochasticity played the leading role in bacterial community assembly throughout the reclamation chronosequence (Fig. 4a). However, an obvious elevation of determinism in bacterial community assembly was observed after tidal flats reclamation. In contrast, fungal community assembly showed a remarkable shift from stochasticity to determinism after tidal flats reclamation (Fig. 4b). The increased relative contribution of homogeneous selection (from 30.9% in DT to 43.27% in ST3) further explained the up-regulating role of determinism in shaping bacterial community assembly (Fig. 4c). Also, the substantially increased relative contribution of homogeneous selection (from 16.3% in DT to 59.8% in ST3) caused the transition from stochasticity to determinism in fungal community assembly (Fig. 4d).

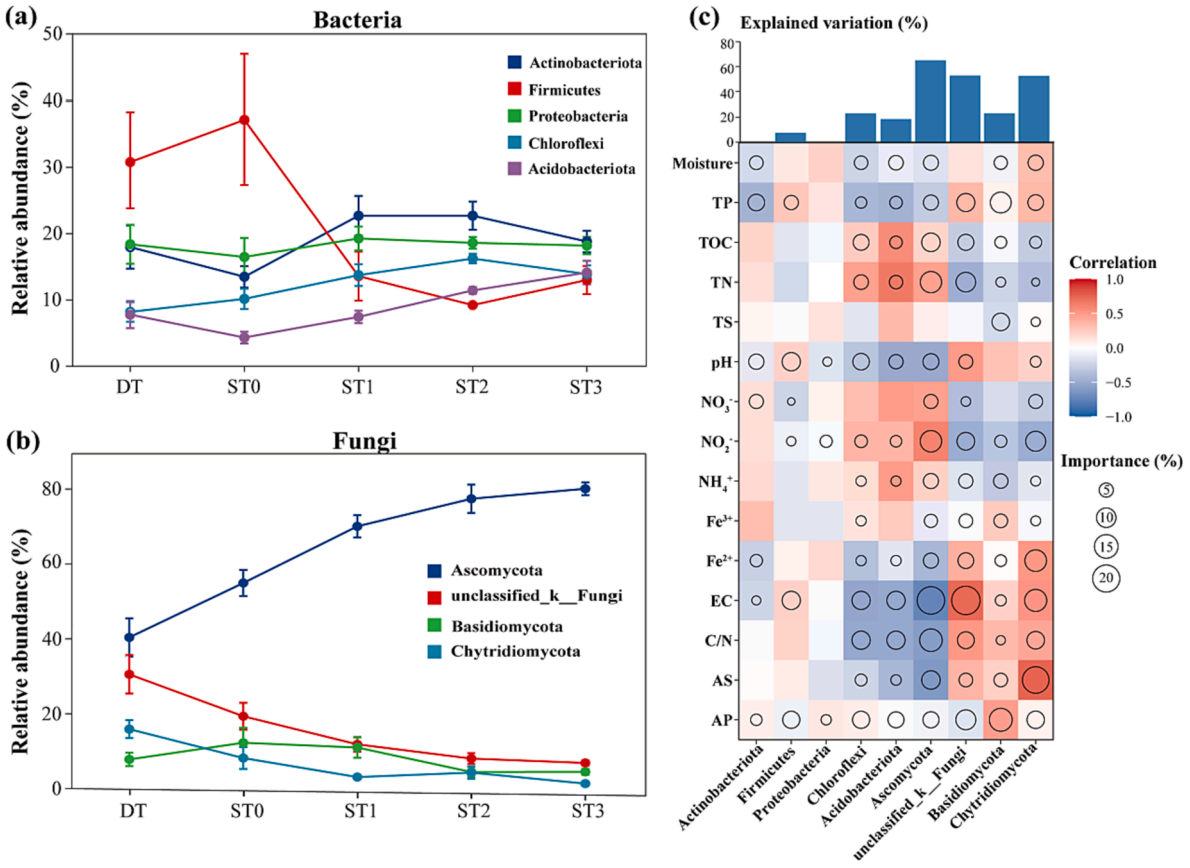


Fig. 2. Responses of microbes to coastal reclamation. Taxonomic composition of bacterial (a) and fungal (b) community at the phylum level. Only the phyla with average abundance > 5% are shown. (c) Contributions of soil physicochemical properties to the relative abundance of bacterial and fungal phyla based on correlation and random forest model. Circle size represents the variables' importance (i.e., percentage of increase of mean square error calculated via random forest model). Colors represent Spearman's correlations.

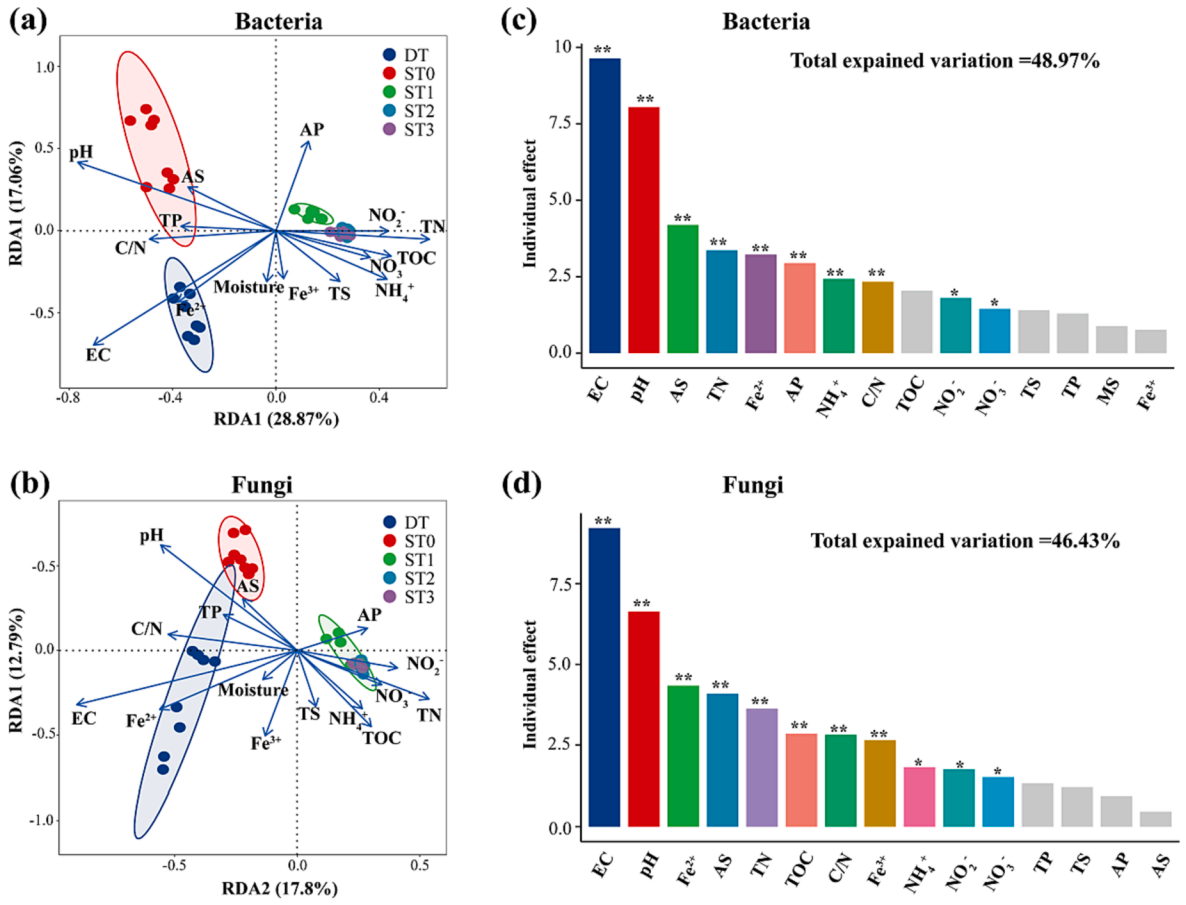


Fig. 3. Variation in soil physicochemical properties induced by coastal reclamation and their impacts on microbial communities. Distance-based redundancy analysis (dbRDA) show factors impacting bacterial (a) and fungal (b) community. The percentage in parentheses represents the variation explained by each axis. (c) Individual impact of each soil property is calculated based on rdacca.hp package. * represents $p < 0.05$; ** represents $p < 0.01$; *** represents $p < 0.001$.

In addition to revealing the assembly mechanisms of entire community, iCAMP can provide useful information on the relative importance of different ecological processes in individual phylogenetic bins or groups. For this purpose, the 14,047 and 7394 observed OTUs of the bacterial and fungal communities were divided into 271 and 145 phylogenetic bins, respectively (Table S5 and S6). Given that homogeneous selection was the key ecological process regulating bacterial and fungal assembly after tidal flats reclamation, bins containing bacteria and fungi contributing to the reclamation-induced changes of homogeneous selection were prioritized. In terms of bacterial community, homogeneous selection dominated 65 bins (23.98% of bin numbers and 26.28% relative abundance) for DT, 64 bins (23.61% of bin numbers and 43.62% relative abundance) for ST0, 88 bins (32.47% of bin numbers and 53.95% relative abundance) for ST1, 91 bins (33.58% of bin numbers and 54.81% relative abundance) for ST2, and 89 bins (32.84% of bin numbers and 56.91% relative abundance) for ST3 (Table S5). Two of the major bins were *Firmicutes* (Bin 209, 16.35% in total abundance of bins) and *Actinobacteriota* (Bin 228, 7.71% in total abundance of bins) (Fig. S9a). *Proteobacteria*, *Acidobacteriota*, *Firmicutes*, and *Actinobacteriota* were further found to contribute much more of the reclamation-induced increases in homogeneous selection of bacterial community (Fig. S9b).

Regarding fungal community, homogeneous selection dominated 14 bins (9.66% of bin numbers and 34.50% relative abundance) for DT, 16 bins (11.03% of bin numbers and 31.31% relative abundance) for ST0, 19 bins (13.10% of bin numbers and 41.06% relative abundance) for ST1, 23 bins (15.86% of bin numbers and 45.43% relative abundance) for ST2, and 20 bins (32.84% of bin numbers and 38.09% relative abundance) for ST3 (Table S6). The major bins were *Ascomycota* (Bin 125, 21.21% in total abundance of bins) (Fig. S10a). *Ascomycota* were further found to contribute much more of the reclamation-induced increases in homogeneous selection of fungal community (Fig. S10b).

Random forest analysis further suggested that changes in soil EC were the best predictor of homogeneous selection in bacterial and fungal community assembly (Fig. 4e and S11). Furthermore, the community-level habitat niche breadth (*Bcom*) was calculated. Bacteria had a much higher *Bcom* value than fungi throughout the reclamation chronosequence (Fig. S11c). Reclamation of tidal flats to paddy soils slightly increased the niche breadths of bacterial community except for a slight decrease in ST0 ($p > 0.05$), while significantly decreased those of fungal community (Fig. 4f and g).

3.5. Microbial co-occurrence networks along the reclamation chronosequence

Bacterial and fungal co-occurrence networks were created to evaluate microbial coexistence relationships along the reclamation chronosequence. Compared to tidal flats, bacterial and fungal co-occurrence networks in reclaimed soils were less complex, and associations were less connected, as evidenced by the multiple network properties (i.e., nodes, links, degree, centrality, and linkage density) (Fig. 5a-d). These findings indicated that reclamation of tidal flats to paddy soils reduced the complexity of bacterial and fungal networks. Furthermore, the impacts of this land-use change on bacterial and fungal co-occurrence networks were tightly related to community composition and responsive species. For example, variations in degree and linkage density following reclamation of tidal flats to paddy soils were remarkably and positively tied to the Bray-Curtis dissimilarities of soil microbiomes

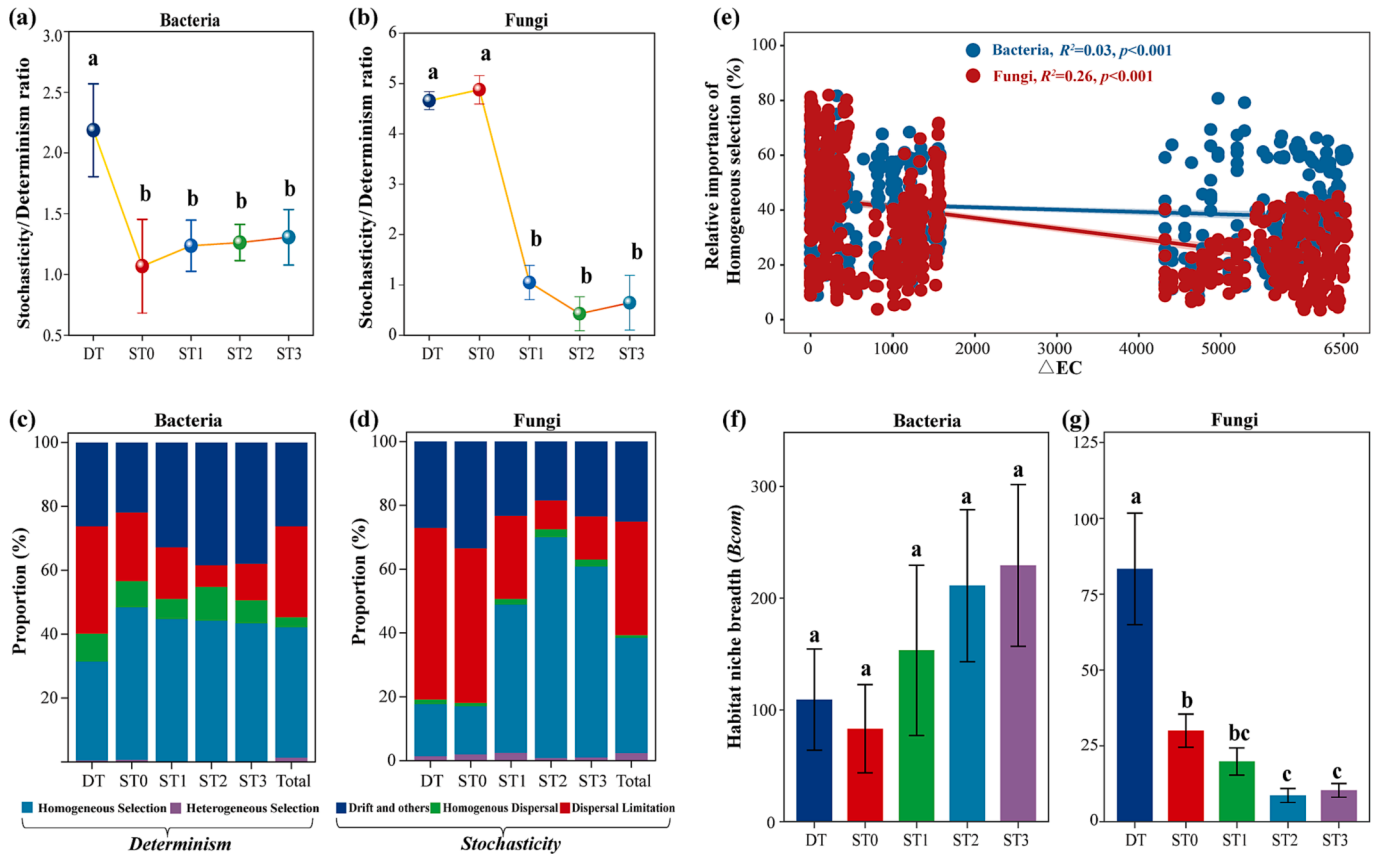


Fig. 4. Ecological assembly mechanisms of microbial communities. Stochasticity/determinism ratio in bacterial (a) and fungal (b) community assembly. Contributions of ecological assembly processes governing bacterial (c) and fungi (d) community assembly. (e) Relationships of the relative contribution of homogeneous selection to variations in soil EC following coastal reclamation. Habitat niche breadth (*Bcom*) of bacterial (f) and fungal (g) community. Different lowercase letters indicate significance between sites at $p < 0.05$.

(Fig. S12). The proportion and average degree of responsive species in the networks, such as *Firmicutes, Ascomycota, unclassified_k_Fungi*, and *Chytridiomycota* phyla showed similar trends to their relative abundance along the reclamation chronoequence (Fig. S13).

Robustness and vulnerability indices were calculated to evaluate the stability of co-occurrence networks (Fig. 5e and f). Both bacterial and fungal co-occurrence networks in tidal flats had a higher robustness but lower vulnerability than those in reclamation soils. These results showed that reclamation of tidal flats to paddy soils decreased the stability of bacterial and fungal networks.

3.6. Linking microbial diversity, network complexity, and multifunctionality

Reclamation of tidal flats to paddy soils generally reduced the multifunctionality of coastal ecosystems, although it gradually increased with increasing years of reclamation (Fig. 6a). Random forest analysis showed that fungal network complexity and fungal α-diversity were two main drivers of multifunctionality (Fig. 6b). Multifunctionality variation patterns observed were in line with the variation patterns of fungal network complexity (fungal network complexity index, degree, linkage density, and eigenvector centrality) and α-diversity (fungal α-diversity index, richness, and Shannon index) (Fig. 6c, d and S14). However, bacteria related variables only had non-significant associations with multifunctionality (Fig. S14).

Structural equation modeling was further performed to examine whether the close associations between multifunctionality and fungal network complexity and α-diversity were maintained when multiple drivers of multifunctionality (such as bacterial community properties [bacterial network complexity and α-diversity] and soil physicochemical properties [soil EC, pH and MS]) were considered simultaneously (Fig. S2, a priori model). To simplify the structural equation modeling, we did not include microbial community composition in the modeling as they contributed less than α-diversity to soil multifunctionality (Fig. 6b). Structural equation modeling showed that the changes in multifunctionality were still largely influenced by reclamation-induced changes in fungal network complexity and α-diversity even after accounting for other key factors (Fig. 6e). These results suggested the decline in fungal network complexity and α-diversity induced by land reclamation contributed to the decrease of coastal ecosystem multifunctionality.

4. Discussion

4.1. Tidal flats reclamation results in contrasting α-diversity response patterns and causes biotic homogenization of soil microbiomes

Land reclamation is an important contributor to biodiversity loss and functioning deterioration in coastal ecosystems (Wang et al., 2023; Wang et al., 2021b). To date, several studies have been performed to uncover the influences of coastal reclamation on soil microbiomes (Ding et al., 2017; Cui et al.,2012; Wang et al., 2021a). However, most of these studies focused on soil bacterial community. Recently, fungi have also been found to play an important role in coastal ecosystem functioning (e.g., N transformations, N_2O production and multifunctionality) (Li et al., 2022a; Sun et al., 2023; Su et al., 2021a). Therefore, to fully understand the impacts of land reclamation on soil microbiomes and ecosystem function in coastal regions, fungal community should be

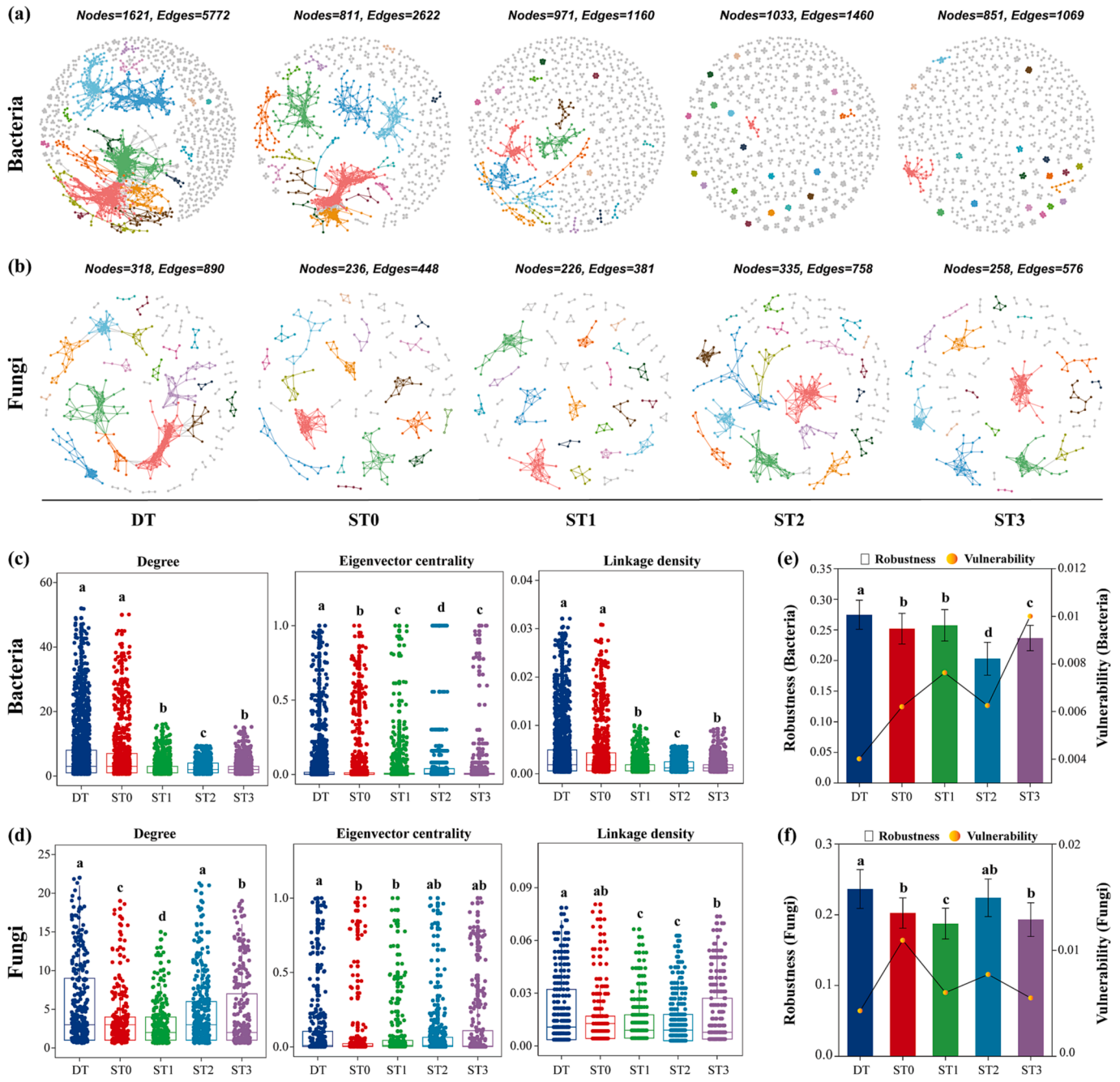


Fig. 5. Microbial co-occurrence patterns along the reclamation chronosequence. Bacterial (a) and fungal (b) co-occurrence networks. The top 18 modules with the number of nodes are shown in different colors, and other modules are shown in grey. Multiple network properties of bacterial (c) and fungal (d) co-occurrence networks. Stability of bacterial (e) and fungal (f) co-occurrence networks. Robustness measured as the proportion of taxa remained with 50% of the taxa randomly removed from each of network. Vulnerability measured by maximum node vulnerability in each network. Different lowercase letters in boxplots indicate significance between sites at $p < 0.05$.

considered. Here we present, to our knowledge, the first study to investigate microbial community succession incorporating both bacteria and fungi along a reclamation chronosequence in coastal regions.

The strikingly different α-diversity patterns of bacterial and fungal communities were observed after tidal flats reclamation (Fig. 1a and b). Bacterial α-diversity significantly increased (except that in ST0), whereas fungal α-diversity decreased. The increase in bacterial α-diversity was identical with previous studies of coastal reclamation to paddy soils (Cui et al., 2012; Ding et al., 2017; Wang et al., 2021a), corn fields (Wang et al., 2020), and wheat fields (Hua et al., 2017). Random forest analysis showed that soil pH, EC, and NO_x^- were more important factors for predicting bacterial α-diversity (Fig. S5a-c). The close relationship between bacterial α-diversity and soil pH may be due to narrow pH ranges for optimal growth of bacteria (Rousk et al., 2010). A maximum decrease of 0.28 pH unit in this work after tidal flats reclamation may therefore help some bacterial taxa to relieve high alkaline pressure and increase bacterial α-diversity. This was further supported by the significant decrease in bacterial α-diversity in ST0, but accompanied by a remarkable increase in pH (Fig. S3 and S4). Despite the importance of pH, the reduced EC and increased NO_x^- were also significant contributors to the increased bacterial α-diversity (Fig. S5c). This was confirmed by previous studies showing that reduced salinity stress and improved nutrient availability increased bacterial α-diversity following coastal reclamation to paddy soils (Ding et al., 2017; Wang

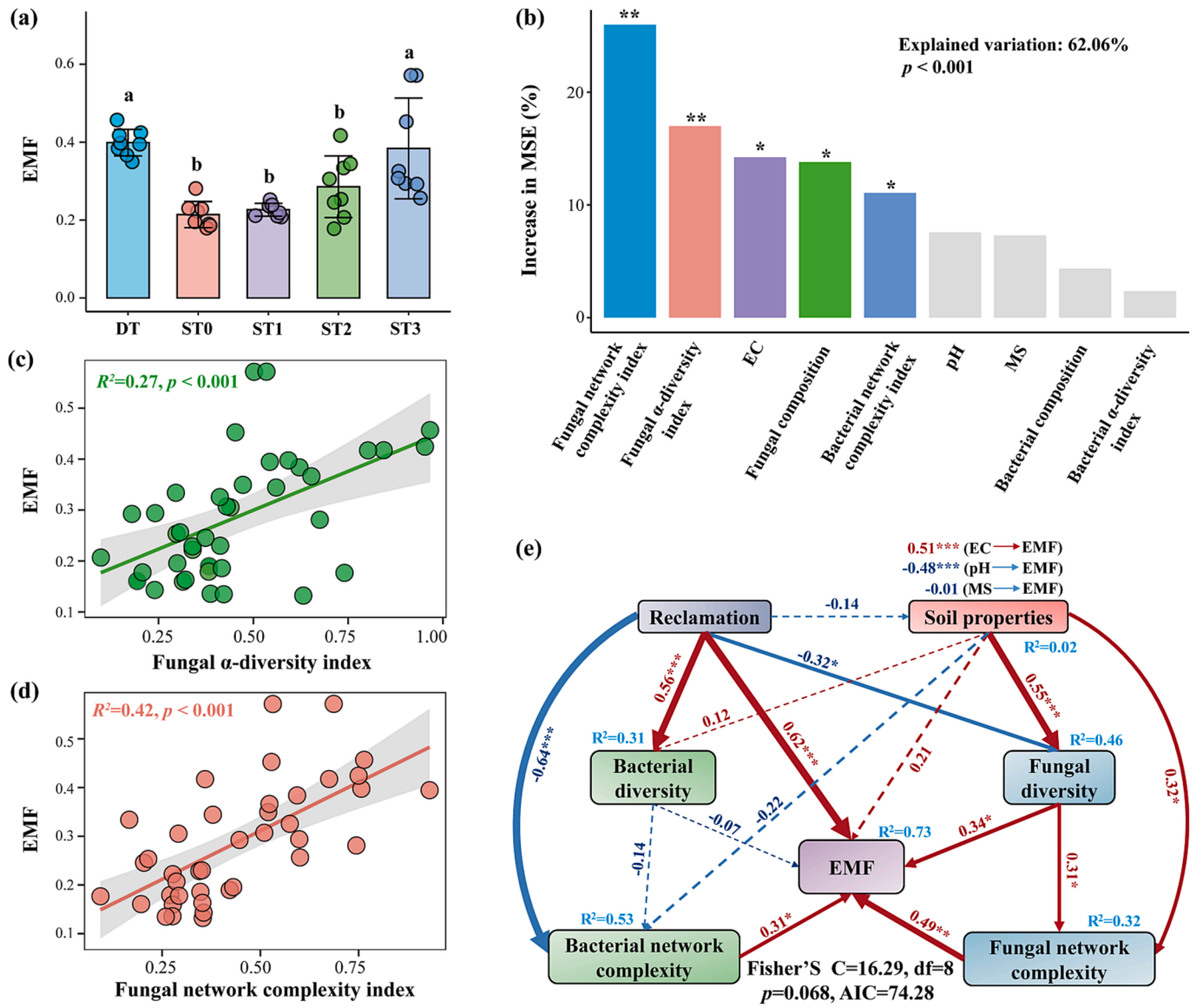


Fig. 6. Drivers of coastal ecosystem multifunctionality (EMF). (a) Variations in coastal ecosystem multifunctionality along the reclamation chronosequence. (b) Random forest indicating the potential drivers of multifunctionality. Percentage increases in the MSE (mean squared error) of variables was used to estimate the importance of these predictors, and higher MSE% values implied more important predictors. Bacterial and fungal community composition are represented by the first axis of principal coordinates analysis. (c) The relationships between multifunctionality and fungal α-diversity index. (d) The relationships between multifunctionality and fungal network complexity index. (e) Structural equation modeling describing the direct and indirect effects of reclamation-induced changes in soil physicochemical properties (EC, pH and MS), microbial diversity (α-diversity index) and network complexity (network complexity index) on multifunctionality. Numbers adjacent to arrows show standardized path coefficients. We remove the relationship between reclamation and fungal network complexity with a coefficient close to zero to improve its overall goodness-of-fit. Blue and red arrows indicate negative and positive relationships, respectively. Continuous and dashed arrows indicate significant and no-significant relationships, respectively. The R^2 values indicate the proportion of variance explained. * represents $p < 0.05$; ** represents $p < 0.01$; *** represents $p < 0.001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al., 2021a). Regarding the fungal α-diversity, the reduced EC and Fe^{2+} were identified as the key factors contributing to the decline in fungal α-diversity after tidal flats reclamation (Fig. S5f). The critical role of EC in regulating fungal α-diversity may be due to the high-salinity niche preference of specific fungal species in coastal regions (Coleine et al., 2022; Chen et al., 2022c). The importance of Fe^{2+} in influencing fungal α-diversity may be because iron is essential for multiple metabolic reactions in fungal cells (Wang et al., 2018).

β-diversity, a crucial aspect of microbial community, is often overlooked in the study of microbial succession (Dini-Andreote et al., 2015; Yang et al., 2022a). In this work, a significant decrease in bacterial and fungal β-diversity was observed along the reclamation chronosequence (Fig. 1e and f). This may be due to the fact that reclamation of tidal flats to paddy soils reduced the niche heterogeneity by homogenizing resources (e.g., EC and pH) (Zhu et al., 2021; Huang et al., 2021). The homogenized resources were discovered to diminish the variation in β-diversity of soil microbiomes (Kong et al., 2022; Jiao et al., 2018). Moreover, the relative contribution of homogeneous selection to bacterial and fungal community assembly obviously increased after tidal flats reclamation (Fig. 4c and d), which may ultimately reduce β-diversity (Kong et al., 2022; Xu et al., 2021). These explanations could be confirmed by the strong dependence of β-diversity on EC and pH (Table S4) and by the remarkable associations between community composition and potential biotic interactions based on co-occurrence networks (Fig. S12) (Kong et al., 2022; Zhang et al., 2021c).

Our results showed that reclamation of tidal flats to paddy soils

significantly decreased the relative abundance of the bacterial phylum *Firmicutes* and taxa affiliated with this phylum, e.g., *Planococcaceae* and *Exiguobacteraceae* (Fig. 2a and S8a). Given these taxa have a high-salinity niche preference (Zhang et al., 2021a; Ebrahimipour et al., 2014; Kim et al., 2005), this alteration was most likely caused by the decrease in soil salinity. With the exception of *unclassified_k_Fungi, Ascomycota*, and *Chytridiomycota* constituted the dominant fungal phylum in tidal flats (Fig. 2b). This was consistent with two recent studies in coastal regions (Su et al., 2021b; Sen et al., 2023). However, their relative abundance polarized along the reclamation chronosequence (Fig. 2b). Tidal flats reclamation remarkably increased the relative abundance of the fungal phylum *Ascomycota*, and the fungal family *Pseudeurotiaceae* was the major contributor (Fig. 2b and S8c). *Ascomycota* and *Pseudeurotiaceae* have a high affinity for nutrients, which have been found to be significantly enriched with N amendments (Guo et al., 2019; Ji et al., 2020). Random forest and correlation analyses confirmed that nitrogen components (TN, NO_3^--N and NO_2^--N) were important predictors for predicting the relative abundance of *Ascomycota* and *Pseudeurotiaceae* (Fig. 2c and S8d). The significantly decreased relative abundance of the fungal phylum *Chytridiomycota* may be due to the decline in salinity (indicated by EC), as its relative abundance was positively associated with EC (Fig. 2c). This explanation was supported by a previous study, which reported that the relative abundance of *Chytridiomycota* significantly increased along a salinity gradient (Mohamed and Martiny, 2011). Overall, our results suggested that reclamation of tidal flats to paddy soils resulted in contrasting α-diversity patterns of bacterial and fungal communities but induced biotic homogenization of them, which may weaken or even lose certain soil ecological functions (Gao et al., 2022; Mori et al., 2018).

4.2. Tidal flats reclamation enhances determinism in community assembly

Elucidating the fundamental assembly mechanisms and drivers that underpin soil microbial community succession is essential for understanding and predicting the response of ecosystem functioning to environmental change (Zhou and Ning, 2017; Liu et al., 2021). While there have been increasing reports on the relative importance of deterministic and stochastic processes mediating microbial community succession, evidence from coastal ecosystems, particularly following disturbances (e.g., land-use change) remains minimal. In this work, we disentangled the ecological mechanisms governing microbial community succession along a reclamation chronosequence in coastal ecosystems. Stochastic processes were found to govern both bacterial and fungal community assembly in tidal flats (Fig. 4a-d), and this agreed to earlier studies (Huang et al., 2022; Zhao et al., 2022c). However, two distinct assembly patterns were observed after tidal flats reclamation, i.e., overarching stochasticity accompanied by an increase in determinism (homogeneous selection) for bacterial community assembly, and determinism (homogeneous selection) exceeding the stochasticity for fungal community assembly (Fig. 4a-d). The different assembly patterns between bacterial and fungal communities can be explained by the size-plasticity hypothesis, i.e., larger organisms (e.g., fungi) are more affected by environmental filtering than smaller organisms (e.g., bacteria) (Huang et al., 2022; Huang et al., 2021). The significantly wider habitat niche breadths of bacteria than fungi throughout the reclamation chronosequence can further support this explanation (Fig. 4 and S11c). Organisms with a wider niche breadth were reported to be more metabolically flexible and be less impacted by deterministic processes (Chen et al., 2022d; Chen et al., 2021).

Bacterial community assembly pattern was consistent with that in traditional paddy agroecosystem succession (Liu et al., 2020b). However, the driving factors were different. Soil SOC was identified as the key factor promoting deterministic assembly of bacterial community in the development of traditional paddy soils (Liu et al., 2020b; Jiao and Lu, 2020). However, in this work, soil EC was found to be more important in up-regulating the role of determinism in microbial assembly (Fig. 4e and S11). This may be due to the different microbial source as the dominant species in tidal flats had a high-salinity niche preference (e.g., *Planococcaceae*, *Exiguobacteraceae*, and *Chytridiomycota*). Overall, our study demonstrated that reclamation of tidal flats to paddy soils up-regulated the role of determinism in microbial community assembly over time, especially in fungal community assembly. This may be detrimental to maintaining ecosystem functioning if the species selected by deterministic processes are those that influence ecosystem functioning (Mori et al., 2018; Zhang et al., 2022).

4.3. Tidal flats reclamation simplifies and destabilizes microbial networks

Uncovering the structure of soil microbial coexistence is crucial for gaining a better understanding of how soil ecological functions evolve in response to environmental change (Gao et al., 2022). The impacts of coastal reclamation on soil microbial coexistence have long been overlooked. Here, we found reclamation of tidal flats to paddy soils simplified microbial networks and consequently reduced network stability (Fig. 5). These were in line with previous studies reporting that land-use type conversion reduced microbial network complexity and/or stability in floodplain (Yu et al., 2022), cropland (Kong et al., 2022), and grassland ecosystems (Luo et al., 2023). Reclamation of tidal flats to paddy soils strongly changed soil physicochemical properties, especially salinity (indicated by EC) in this work, which can exert a strong selective pressure on the soil microbial communities (Fig. 4e). The selective pressure can further exclude more species from the networks, resulting in simpler and less stable networks (Gao et al., 2022; Kong et al., 2022; Ratzke et al., 2020). This point could be supported by the fact that more species were replaced as the EC changed following reclamation of coastal wetlands to paddy soils (Fig. S6 and Table S4). In addition, the reduced proportion and connectivity of some key phyla such as *Firmicutes* and *Chytridiomycota* in the co-occurrence networks also supported this explanation (Fig. S13a-d), as these phyla were strongly associated with EC (Fig. 2c). The importance of salinity in maintaining the complexity and stability of microbial networks has also been demonstrated in lakes (Ji et al., 2019), wastewater treatment (Zhao et al., 2022b), and agricultural ecosystems (Ya et al., 2021). Another possible explanation for the simplified and destabilized networks is biotic homogenization following tidal flats reclamation to paddy soils. When biotic homogenization happens, microbes within a community become more similar and compete for similar resources, leading to heightened competition within the community (Gao et al., 2022; Olden and Poff, 2004). In general, reclamation of tidal flats to paddy soils decreased the interaction strength between microbial taxa and consequently resulted in more unstable networks. It will be difficult to restore and maintain normal ecological functions when the networks collapse.

4.4. Fungal network complexity and community diversity determine the changes in multifunctionality along the reclamation chronosequence

Coastal wetlands provide multiple ecosystem functions but are extremely vulnerable to land reclamation (Wang et al., 2023; Wang et al., 2021b). Existing studies have typically concentrated on how land reclamation affects certain ecological functions in coastal wetlands, such as C and N transformations (Jiang et al., 2021; Tan et al., 2022; Tan et al., 2020) and greenhouse gas emissions (Tan et al., 2020; Yang et al., 2022b). However, the influences of coastal reclamation on multiple ecological functions (i.e., multifunctionality) are largely unknown. Here, we provide evidence that reclamation of tidal flats to paddy soils overall reduced soil multifunctionality in a 53-year reclamation chronoseuqence (Fig. 6a). Two key drivers were fungal α-diversity and network complexity (Fig. 6b-e). Relationships between soil microbial diversity and ecosystem multifunctionality were context-dependent (Hu et al., 2021; Han et al., 2022a). In this work, fungal rather than bacterial α-diversity dominated the variations in soil multifunctionality in the

reclamation chronosequence, possibly because bacteria and fungi were involved in different resource partitioning processes in coastal ecosystems (Li et al., 2022a). In coastal ecosystems, fungi initially decompose organic matter by extracellular enzyme activities and physical destruction (Buchan et al., 2003; Li et al., 2022a). As decomposition progresses, the macromolecular organic matter becomes smaller, and then bacteria replace fungi and become increasingly vital in the later stages of decomposition (Li et al., 2022a). Therefore, the reduced fungal α-diversity following tidal flats reclamation may limit the nutrient availability, disrupt biogeochemical cycles, and subsequently reduce soil multifunctionality. This point was further supported by the positive and significant relationships between fungal α-diversity and extracellular enzyme activities in the reclamation chronosequence (Fig. S15a). Notably, soil multifunctionality gradually increased with increasing years of reclamation (Fig. 6a), as did the relative abundance of functional genes related to C, N, S, and P cycles (Fig. S3 and Table S2). Therefore, in the context of coastal reclamation, whether bacteria can compensate for the loss of fungal functions due to habitat changes needs to be further explored.

Despite the importance of fungal α-diversity, our work indicated that fungal network complexity also played an important role in driving soil multifunctionality along the reclamation chonosequence (Fig. 6b and e). This agreed to earlier studies suggesting that biota with simpler interactions contributed less to ecosystem functioning (Chen et al., 2022e; Morrien et al., 2017; Wagg et al., 2019) and anthropogenic disturbances reduced ecosystem functioning through simplifying ecological communities (Luo et al., 2023; Gossner et al., 2016). When comparing the relative importance of the two main drivers in shaping soil multifunctionality, fungal network complexity was found to be relatively more important (Fig. 6e and Table S7). This result implied that the ecosystem functioning may be dependent more on the ecological interactions between taxa than on the number of taxa (Luo et al., 2023; Chen et al., 2022e). However, in this work, we highlight that fungal network complexity and diversity jointly shaped the changes in soil multifunctionality in the reclamation chronosequnce, as α-diversity is the basis for the complex microbial interactions and did play an important role in multifunctionality (Table S7). Overall, these findings suggest that the reduction in fungal network complexity and community diversity induced by land reclamation are detrimental to the maintenance of coastal ecosystem multifunctionality.

5. Conclusions

This study provides a comprehensive insight into the influences of land reclamation on coastal soil microbiomes using a 53-year chronosequence of paddy soil following reclamation from tidal flats. Our findings reveal that tidal flats reclamation to paddy soils caused contrasting α-diversity response patterns of bacterial and fungal communities, and decreased both of their β-diversity, while increased the importance of homogeneous selection in their assembly. Tidal flats reclamation to paddy soils also reduced the complexity and robustness of the co-occurrence networks, as well as the multifunctionality of coastal ecosystems. Notably, fungal network complexity and α-diversity were two main drivers of the decreased multifunctionality. This highlights the need to improve the complexity of soil fungal community to restore coastal ecosystem functions deteriorated by land reclamation. Overall, these findings have advanced the current understanding on the influences of coastal reclamation on soil microbiomes and the multiple ecological functions they support, informing land management in coastal ecosystems under global change.

CRediT authorship contribution statement

Cheng Chen: Investigation, Data curation, Writing – original draft. **Guoyu Yin:** Conceptualization, Writing – review & editing, Funding acquisition. **Lijun Hou:** Funding acquisition. **Yinghui Jiang:** Investigation. **Dongyao Sun:** Investigation. **Xia Liang:** Investigation. **Ping Han:** Investigation. **Yanling Zheng:** Software. **Min Liu:** Conceptualization, Writing – review & editing, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2023.108151.

References

Aslani, F., Geisen, S., Ning, D., Tedersoo, L., Bahram, M., de Vries, F., 2022. Towards revealing the global diversity and community assembly of soil eukaryotes. Ecol. Lett. 25 (1), 65–76.
Banerjee, S., van der Heijden, M.G.A., 2023. Soil microbiomes and one health. Nat. Rev. Microbiol. 21 (1), 6–20.
Blum, M.D., Roberts, H.H., 2009. Drowning of the Mississippi Delta due to insufficient sediment supply and global sea-level rise. Nat. Geosci. 2, 488–491.
Buchan, A., Newell, S.Y., Butler, M., Biers, E.J., Hollibaugh, J.T., Moran, M.A., 2003. Dynamics of bacterial and fungal communities on decaying salt marsh grass. Appl. Environ. Microbiol. 69 (11), 6676–6687.
Caporaso, J.G., Lauber, C.L., Walters, W.A., Berg-Lyons, D., Lozupone, C.A., Turnbaugh, P.J., Fierer, N., Knight, R., 2011. Global patterns of 16S rRNA diversity at a depth of millions of sequences per sample. PNAS 108 (supplement_1), 4516–4522.
Chen, Q.-L., Hu, H.-W., Yan, Z.-Z., Li, C.-Y., Nguyen, B.-A., Sun, A.-Q., Zhu, Y.-G., He, J.-Z., 2021. Deterministic selection dominates microbial community assembly in termite mounds. Soil Biol. Biochem. 152, 108073.
Chen, Q.-L., Hu, H.-W., Sun, A.-Q., Zhu, Y.-G., He, J.-Z., 2022d. Aridity decreases soil protistan network complexity and stability. Soil Biol. Biochem. 166, 108575.
Chen, C., Li, Y.e., Yin, G., Hou, L., Liu, M., Jiang, Y., Zheng, D., Wu, H., Zheng, Y., Sun, D., 2022a. Antibiotics sulfamethoxazole alter nitrous oxide production and pathways in estuarine sediments: Evidenced by the N^{15}–O^{18} isotopes tracing. J. Hazard. Mater. 437, 129281.
Chen, H., Ma, K., Huang, Y.u., Fu, Q.i., Qiu, Y., Yao, Z., 2022c. Significant response of microbial community to increased salinity across wetland ecosystems. Geoderma 415, 115778.
Chen, C., Pan, J., Xiao, S., Wang, J., Gong, X., Yin, G., Hou, L., Liu, M., Zheng, Y., 2022b. Microplastics alter nitrous oxide production and pathways through affecting microbiome in estuarine sediments. Water Res. 221, 118733.
Chen, W., Wang, J., Chen, X., Meng, Z., Xu, R., Duoji, D., Zhang, J., He, J., Wang, Z., Chen, J., Liu, K., Hu, T., Zhang, Y., 2022e. Soil microbial network complexity predicts ecosystem function along elevation gradients on the Tibetan Plateau. Soil Biol. Biochem. 172, 108766.
Churka Blum, S., Lehmann, J., Solomon, D., Caires, E.F., Alleoni, L.R.F., 2013. Sulfur forms in organic substrates affecting S mineralization in soil. Geoderma 200–201, 156–164.
Coleine, C., Stajich, J.E., Selbmann, L., 2022. Fungi are key players in extreme ecosystems. Trends Ecol. Evol. 37 (6), 517–528.
Cui, J., Meng, H., Nie, M., Chen, X., Li, Z., Bu, N., Li, B., Chen, J., Quan, Z., Fang, C., 2012. Bacterial succession during 500 years of soil development under agricultural use. Ecol. Res. 27, 793–807.
de Vries, F.T., Griffiths, R.I., Bailey, M., Craig, H., Girlanda, M., Gweon, H.S., Hallin, S., Kaisermann, A., Keith, A.M., Kretzschmar, M., Lemanceau, P., Lumini, E., Mason, K. E., Oliver, A., Ostle, N., Prosser, J.I., Thion, C., Thomson, B., Bardgett, R.D., 2018. Soil bacterial networks are less stable under drought than fungal networks. Nat. Commun. 9, 3033.

Ding, L., Su, J., Li, H., Zhu, Y., Cao, Z., 2017. Bacterial succession along a long-term chronosequence of paddy soil in the Yangtze River Delta, China. Soil Biol. Biochem. 104, 59–67.
Dini-Andreote, F., Stegen, J.C., van Elsas, J.D., Salles, J.F., 2015. Disentangling mechanisms that mediate the balance between stochastic and deterministic processes in microbial succession. PNAS 112 (11).
Ebrahimipour, G., Gilavand, F., Karkhane, M., Kavyanifard, A.A., Teymouri, M., Marzban, A., 2014. Bioemulsification activity assessment of an indigenous strain of halotolerant Planococcus and partial characterization of produced biosurfactants. Int. J. Environ. Sci. Technol. 11 (5), 1379–1386.
Edgar, R.C., 2013. UPARSE: highly accurate OTU sequences from microbial amplicon reads. Nat. Methods 10 (10), 996–998.
Edgar, R.C., Haas, B.J., Clemente, J.C., Quince, C., Knight, R., 2011. UCHIME improves sensitivity and speed of chimera detection. Bioinformatics 27, 2194-2200.
Fanin, N., Gundale, M.J., Farrell, M., Ciobanu, M., Baldock, J.A., Nilsson, M.C., Kardol, P., Wardle, D.A., 2018. Consistent effects of biodiversity loss on multifunctionality across contrasting ecosystems. Nat. Ecol. Evol. 2, 269–278.
Fluet-Chouinard, E., Stocker, B.D., Zhang, Z., Malhotra, A., Melton, J.R., Poulter, B., Kaplan, J.O., Goldewijk, K.K., Siebert, S., Minayeva, T., Hugelius, G., Joosten, H., Barthelmes, A., Prigent, C., Aires, F., Hoyt, A.M., Davidson, N., Finlayson, C.M., Lehner, B., Jackson, R.B., McIntyre, P.B., 2023. Extensive global wetland loss over the past three centuries. Nature 614 (7947), 281–286.
Gao, D., Li, X., Lin, X., Wu, D., Jin, B., Huang, Y., Liu, M., Chen, X., 2017. Soil dissimilatory nitrate reduction processes in the Spartina alterniflora invasion chronosequences of a coastal wetland of southeastern China: Dynamics and environmental implications. Plant and Soil 421 (1-2), 383–399.
Gao, D., Hou, L., Liu, M., Li, X., Zheng, Y., Yin, G., Wu, D., Yang, Y.i., Han, P., Liang, X., Dong, H., 2020. Mechanisms responsible for N_2O emissions from intertidal soils of the Yangtze Estuary. Sci. Total Environ. 716, 137073.
Gao, G., Li, H., Shi, Y., Yang, T., Gao, C., Fan, K., Zhang, Y., Zhu, Y., Delgado-Baquerizo, M., Zheng, H., Chu, H., 2022. Continental-scale plant invasions reshuffle the soil microbiome of blue carbon ecosystems. Glob. Chang. Biol. 28, 4423–4438.
Gossner, M.M., Lewinsohn, T.M., Kahl, T., Grassein, F., Boch, S., Prati, D., Birkhofer, K., Renner, S.C., Sikorski, J., Wubet, T., Arndt, H., Baumgartner, V., Blaser, S., Blüthgen, N., Börschig, C., Buscot, F., Diekötter, T., Jorge, L.R., Jung, K., Keyel, A.C., Klein, A.-M., Klemmer, S., Krauss, J., Lange, M., Müller, J., Overmann, J., Pašalić, E., Penone, C., Perović, D.J., Purschke, O., Schall, P., Socher, S.A., Sonnemann, I., Tschapka, M., Tscharntke, T., Türke, M., Venter, P.C., Weiner, C.N., Werner, M., Wolters, V., Wurst, S., Westphal, C., Fischer, M., Weisser, W.W., Allan, E., 2016. Land-use intensification causes multitrophic homogenization of grassland communities. Nature 540 (7632), 266–269.
Guo, Q., Yan, L., Korpelainen, H., Niinemets, Ü., Li, C., 2019. Plant-plant interactions and N fertilization shape soil bacterial and fungal communities. Soil Biol. Biochem. 128, 127–138.
Haese, R.R., Wallmann, K., Dahmke, A., Kretzmann, U., Müller, P.J., Schulz, H.D., 1997. Iron species determination to investigate early diagenetic reactivity in marine sediments. Geochim. Cosmochim. Acta 61 (1), 63–72.
Han, S., Tan, S., Wang, A., Chen, W., Huang, Q., 2022a. Bacterial rather than fungal diversity and community assembly drive soil multifunctionality in a subtropical forest ecosystem. Environ. Microbiol. Rep. 14 (1), 85–95.
Han, Z., Xu, P., Li, Z., Lin, H., Zhu, C., Wang, J., Zou, J., 2022b. Microbial diversity and the abundance of keystone species drive the response of soil multifunctionality to organic substitution and biochar amendment in a tea plantation. Glob. Change Biol. Bioenergy. 14 (4), 481–495.
Hu, W., Ran, J., Dong, L., Du, Q., Ji, M., Yao, S., Sun, Y., Gong, C., Hou, Q., Gong, H., Chen, R., Lu, J., Xie, S., Wang, Z., Huang, H., Li, X., Xiong, J., Xia, R., Wei, M., Zhao, D., Zhang, Y., Li, J., Yang, H., Wang, X., Deng, Y., Sun, Y., Li, H., Zhang, L., Chu, Q., Li, X., Aqeel, M., Manan, A., Akram, M.A., Liu, X., Li, R., Li, F., Hou, C., Liu, J., He, J.S., An, L., Bardgett, R.D., Schmid, B., Deng, J., 2021. Aridity-driven shift in biodiversity-soil multifunctionality relationships. Nat. Commun. 12, 5350.
Hua, J., Feng, Y., Jiang, Q., Bao, X., Yin, Y., 2017. Shift of bacterial community structure along different coastal reclamation histories in Jiangsu. Eastern China. Scientific Reports 7, 10096.
Huang, L., Bai, J., Wang, J., Zhang, G., Wang, W., Wang, X., Zhang, L., Wang, Y., Liu, X., Cui, B., 2022. Different stochastic processes regulate bacterial and fungal community assembly in estuarine wetland soils. Soil Biol. Biochem. 167, 108586.
Huang, X., Wang, J., Dumack, K., Liu, W., Zhang, Q., He, Y., Di, H., Bonkowski, M., Xu, J., Li, Y., 2021. Protists modulate fungal community assembly in paddy soils across climatic zones at the continental scale. Soil Biol. Biochem. 160, 108358.
Ji, M., Kong, W., Yue, L., Wang, J., Deng, Y., Zhu, L., 2019. Salinity reduces bacterial diversity, but increases network complexity in Tibetan Plateau lakes. FEMS Microbiol. Ecol. 95, fiz190.
Ji, L., Ni, K., Wu, Z., Zhang, J., Yi, X., Yang, X., Ling, N., You, Z., Guo, S., Ruan, J., 2020. Effect of organic substitution rates on soil quality and fungal community composition in a tea plantation with long-term fertilization. Biol. Fertil. Soils 56 (5), 633–646.
Jiang, Y., Yin, G., Hou, L., Liu, M., Gao, D., Zhang, Z., Zheng, Y., Han, P., 2021. Variations of dissimilatory nitrate reduction processes along reclamation chronosequences in Chongming Island. China. Soil and Tillage Research 206, 104815.
Jiang, Y., Yin, G., Li, Y.e., Hou, L., Liu, M., Chen, C., Zheng, D., Wu, H., Gao, D., Zheng, Y., Han, P., 2022. Saltwater incursion regulates N_2O emission pathways and potential nitrification and denitrification in intertidal wetland. Biol. Fertil. Soils 59 (5), 541–553.
Jiao, S., Chen, W., Wang, J., Du, N., Li, Q., Wei, G., 2018. Soil microbiomes with distinct assemblies through vertical soil profiles drive the cycling of multiple nutrients in reforested ecosystems. Microbiome 6, 146.
Jiao, S., Lu, Y., 2020. Abundant fungi adapt to broader environmental gradients than rare fungi in agricultural fields. Globle Change Biology 26 (8), 4506–4520.
Jiao, S., Yang, Y., Xu, Y., Zhang, J., Lu, Y., 2020. Balance between community assembly processes mediates species coexistence in agricultural soil microbiomes across eastern China. ISME J. 14 (1), 202–216.
Jiao, S., Qi, J., Jin, C., Liu, Y.u., Wang, Y., Pan, H., Chen, S., Liang, C., Peng, Z., Chen, B., Qian, X., Wei, G., 2022. Core phylotypes enhance the resistance of soil microbiome to environmental changes to maintain multifunctionality in agricultural ecosystems. Glob. Chang. Biol. 28 (22), 6653–6664.
Ju, Z., Du, X., Feng, K., Li, S., Gu, S., Jin, D., Deng, Y., 2021. The Succession of Bacterial Community Attached on Biodegradable Plastic Mulches During the Degradation in Soil. Front. Microbiol. 12, 785737.
Kim, I.G., Lee, M.H., Jung, S.Y., Song, J.J., Oh, T.K., Yoon, J.H., 2005. Exiguobacterium aestuarii sp. nov. and Exiguobacterium marinum sp. nov., isolated from a tidal flat of the Yellow Sea in Korea. Int. J. Syst. Evol. Microbiol. 55, 885–889.
Kong, W., Wei, X., Wu, Y., Shao, M., Zhang, Q., Sadowsky, M.J., Ishii, S., Reich, P.B., Wei, G., Jiao, S., Qiu, L., Liu, L., 2022. Afforestation can lower microbial diversity and functionality in deep soil layers in a semiarid region. Glob. Chang. Biol. 28 (20), 6086–6101.
Kunito, T., Kurita, H., Kumori, M., Sakaguchi, K., Nishizawa, S., Fujita, K., Moro, H., Sawada, K., Miyabara, Y., Toda, H., Nagaoka, K., Ishikawa, Y., 2022. Microbial synthesis of arylsulfatase depends on the soluble and adsorbed sulfate concentration in soils. Eur. J. Soil Biol. 111, 103418.
Lai, J., Zou, Y.i., Zhang, J., Peres-Neto, P.R., 2022. Generalizing hierarchical and variation partitioning in multiple regression and canonical analyses using the rdacca.hp R package. Methods Ecol. Evol. 13 (4), 782–788.
Legendre, P., 2014. Interpreting the replacement and richness difference components of beta diversity. Glob. Ecol. Biogeogr. 23 (11), 1324–1334.
Li, J., Cui, L., Delgado-Baquerizo, M., Wang, J., Zhu, Y., Wang, R., Li, W., Lei, Y., Zhai, X., Zhao, X., Singh, B.K., 2022a. Fungi drive soil multifunctionality in the coastal salt marsh ecosystem. Sci. Total Environ. 818, 151673.
Li, W., Kuzyakov, Y., Zheng, Y., Li, P., Li, G., Liu, M., Alharbi, H.A., Li, Z., 2022b. Depth effects on bacterial community assembly processes in paddy soils. Soil Biol. Biochem. 165, 108517.
Liu, W., Graham, E.B., Zhong, L., Zhang, J., Li, W., Li, Z., Lin, X., Feng, Y., Wang, J., 2020b. Dynamic microbial assembly processes correspond to soil fertility in sustainable paddy agroecosystems. Funct. Ecol. 34 (6), 1244–1256.
Liu, J., Zhu, S., Liu, X., Yao, P., Ge, T., Zhang, X.-H., 2020a. Spatiotemporal dynamics of the archaeal community in coastal sediments: assembly process and co-occurrence relationship. ISME J. 14 (6), 1463–1478.
Liu, L., Zhu, K., Krause, S.M.B., Li, S., Wang, X., Zhang, Z., Shen, M., Yang, Q., Lian, J., Wang, X., Ye, W., Zhang, J., 2021. Changes in assembly processes of soil microbial communities during secondary succession in two subtropical forests. Soil Biol. Biochem. 154, 108144.
Luo, S., Png, G.K., Ostle, N.J., Zhou, H., Hou, X., Luo, C., Quinton, J.N., Schaffner, U., Sweeney, C., Wang, D., Wu, J., Wu, Y., Bardgett, R.D., 2023. Grassland degradation-induced declines in soil fungal complexity reduce fungal community stability and ecosystem multifunctionality. Soil Biol. Biochem. 176, 108865.
Ma, Z., Melville, D.S., Liu, J., Chen, Y., Yang, H., Ren, W., Zhang, Z., Piersma, T., Li, B.o., 2014. Rethinking China's new great wall. Science 346 (6212), 912–914.
Ma, B., Wang, H., Dsouza, M., Lou, J., He, Y., Dai, Z., Brookes, P.C., Xu, J., Gilbert, J.A., 2016. Geographic patterns of co-occurrence network topological features for soil microbiota at continental scale in eastern China. ISME J. 10 (8), 1891–1901.
Mohamed, D.J., Martiny, J.BH., 2011. Patterns of fungal diversity and composition along a salinity gradient. ISME J. 5 (3), 379–388.
Mori, A.S., Isbell, F., Seidl, R., 2018. β-Diversity, Community Assembly, and Ecosystem Functioning. Trends Ecol. Evol. 33 (7), 549–564.
Morrien, E., Hannula, S.E., Snoek, L.B., Helmsing, N.R., Zweers, H., de Hollander, M., Soto, R.L., Bouffaud, M.L., Buee, M., Dimmers, W., Duyts, H., Geisen, S., Girlanda, M., Griffiths, R.I., Jorgensen, H.B., Jensen, J., Plassart, P., Redecker, D., Schmelz, R.M., Schmidt, O., Thomson, B.C., Tisserant, E., Uroz, S., Winding, A., Bailey, M.J., Bonkowski, M., Faber, J.H., Martin, F., Lemanceau, P., de Boer, W., van Veen, J.A., van der Putten, W.H., 2017. Soil networks become more connected and take up more carbon as nature restoration progresses. Nat. Commun. 8, 14349.
Murray, N.J., Phinn, S.R., DeWitt, M., Ferrari, R., Johnston, R., Lyons, M.B., Clinton, N., Thau, D., Fuller, R.A., 2019. The global distribution and trajectory of tidal flats. Nature 565 (7738), 222–225.
Nilsson, R.H., Larsson, K.H., Taylor, A.F.S., Bengtsson-Palme, J., Jeppesen, T.S., Schigel, D., Kennedy, P., Picard, K., Glockner, F.O., Tedersoo, L., Saar, I., Koljalg, U., Abarenkov, K., 2019. The UNITE database for molecular identification of fungi: handling dark taxa and parallel taxonomic classifications. Nucleic Acids Res. 47, D259–D264.
Ning, D., Deng, Y.e., Tiedje, J.M., Zhou, J., 2019. A general framework for quantitatively assessing ecological stochasticity. PNAS 116 (34), 16892–16898.
Ning, D., Yuan, M., Wu, L., Zhang, Y., Guo, X., Zhou, X., Yang, Y., Arkin, A.P., Firestone, M.K., Zhou, J., 2020. A quantitative framework reveals ecological drivers of grassland microbial community assembly in response to warming. Nat. Commun. 11, 4717.
Olden, J.D., Poff, N.L., 2004. Ecological Processes Driving Biotic Homogenization: Testing a Mechanistic Model Using Fish Faunas. Ecology 85 (7), 1867–1875.
Page, A., Miller, R., Keeney, D., 1982. Methods of Soil Analysis, Part 2. Chemical and Microbiological Properties. American Society of Agronomy Inc, Soil Science Society of America Inc, Madison, Wisconsin.

Qiu, L., Zhang, Q., Zhu, H., Reich, P.B., Banerjee, S., van der Heijden, M.G.A., Sadowsky, M.J., Ishii, S., Jia, X., Shao, M., Liu, B., Jiao, H., Li, H., Wei, X., 2021. Erosion reduces soil microbial diversity, network complexity and multifunctionality. ISME J. 15 (8), 2474–2489.
Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., Glockner, F.O., 2013. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. Nucleic Acids Res. 41, D590–D596.
Ratzke, C., Barrere, J., Gore, J., 2020. Strength of species interactions determines biodiversity and stability in microbial communities. Nature Ecology Evolution 4, 376–383.
Rousk, J., Bååth, E., Brookes, P.C., Lauber, C.L., Lozupone, C., Caporaso, J.G., Knight, R., Fierer, N., 2010. Soil bacterial and fungal communities across a pH gradient in an arable soil. ISME J. 4 (10), 1340–1351.
Santos, F., Lima, M.G.M., Espinosa, S., Ahumada, J.A., Jansen, P.A., Spironello, W.R., Hurtado, J., Juen, L., Peres, C.A., 2021. Site and species contribution to beta-diversity in terrestrial mammal communities: Evidence from multiple Neotropical forest sites. Sci. Total Environ. 789, 147946.
Sen, K., Bai, M., Li, J., Ding, X., Sen, B., Wang, G., 2023. Spatial Patterns of Planktonic Fungi Indicate Their Potential Contributions to Biological Carbon Pump and Organic Matter Remineralization in the Water Column of South China Sea. Journal of Fungi 9, 640.
Shipley, B., 2009. Confirmatory path analysis in a generalized multilevel context. Ecology 90 (2), 363–368.
Sommers, L.E., Nelson, D.W., 1972. Determination of Total Phosphorus in Soils: A Rapid Perchloric Acid Digestion Procedure. Soil Sci. Soc. Am. J. 36 (6), 902–904.
Stegen, J.C., Lin, X., Fredrickson, J.K., Chen, X., Kennedy, D.W., Murray, C.J., Rockhold, M.L., Konopka, A., 2013. Quantifying community assembly processes and identifying features that impose them. ISME J. 7 (11), 2069–2079.
Su, X., Wen, T., Wang, Y., Xu, J., Cui, L.i., Zhang, J., Xue, X., Ding, K., Tang, Y., Zhu, Y.-G., 2021a. Stimulation of N_2O emission via bacterial denitrification driven by acidification in estuarine sediments. Glob. Chang. Biol. 27 (21), 5564–5579.
Su, X., Yang, X., Li, H., Wang, H., Wang, Y., Xu, J., Ding, K., Zhu, Y.G., 2021b. Bacterial communities are more sensitive to ocean acidification than fungal communities in estuarine sediments. FEMS Microbiol. Ecol. 97, fiab058.
Sun, K., Cai, J., Liu, X., Yang, L., Li, H., Wang, G., Xu, X., Yu, F., 2023. Effects of nitrogen and phosphorus supply levels and ratios on soil microbial diversity-ecosystem multifunctionality relationships in a coastal nontidal wetland. Sci. Total Environ. 874, 162472.
Sun, C., Zhang, B., Ning, D., Zhang, Y.a., Dai, T., Wu, L., Li, T., Liu, W., Zhou, J., Wen, X., 2021. Seasonal dynamics of the microbial community in two full-scale wastewater treatment plants: Diversity, composition, phylogenetic group based assembly and co-occurrence pattern. Water Res. 200, 117295.
Tan, L., Ge, Z., Zhou, X., Li, S., Li, X., Tang, J., 2020. Conversion of coastal wetlands, riparian wetlands, and peatlands increases greenhouse gas emissions: A global meta-analysis. Globle Change Biology 26 (3), 1638–1653.
Tan, L., Ge, Z., Ji, Y., Lai, D.Y.F., Temmerman, S., Li, S., Li, X., Tang, J., 2022. Land use and land cover changes in coastal and inland wetlands cause soil carbon and nitrogen loss. Glob. Ecol. Biogeogr. 31 (12), 2541–2563.
Wagg, C., Schlaeppi, K., Banerjee, S., Kuramae, E.E., van der Heijden, M.G.A., 2019. Fungal-bacterial diversity and microbiome complexity predict ecosystem functioning. Nat. Commun. 10, 4841.
Wang, Y.-F., Chen, P., Wang, F.-H., Han, W.-X., Qiao, M., Dong, W.-X., Hu, C.-S., Zhu, D., Chu, H.-Y., Zhu, Y.-G., 2022. The ecological clusters of soil organisms drive the ecosystem multifunctionality under long-term fertilization. Environ. Int. 161, 107133.
Wang, R., Creamer, C.A., Wang, X., He, P., Xu, Z., Jiang, Y., 2016. The effects of a 9-year nitrogen and water addition on soil aggregate phosphorus and sulfur availability in a semi-arid grassland. Ecol. Ind. 61, 806–814.
Wang, H., Li, Z., Niu, J., Xu, Y., Ma, L., Lu, A., Wang, X., Qian, Z., Huang, Z., Jin, X., Leng, Q., Wang, J., Zhong, J., Sun, B., Meng, G., 2018. Antiviral effects of ferric ammonium citrate. Cell Discovery 4, 14.
Wang, F., Peng, W., Zhang, B.o., Cao, Y., Zhao, J., Cao, H., 2021a. Succession of bacterial community composition in coastal agricultural soils along a 1000-year reclamation chronosequence in Hangzhou Bay. China. Ecological Indicators 121, 106972.
Wang, X., Yang, J., Xie, X., Chen, X., Pu, L., Zhang, X., 2020. Soil microbial succession with soil development since costal reclamation. Catena 187, 104393.
Wang, X., Xiao, X., Xu, X., Zou, Z., Chen, B., Qin, Y., Zhang, X., Dong, J., Liu, D., Pan, L., Li, B., 2021b. Rebound in China's coastal wetlands following conservation and restoration. Nat. Sustainability 4, 1076–1083.
Wang, X., Xiao, X., Zhang, X.i., Wu, J., Li, B.o., 2023. Rapid and large changes in coastal wetland structure in China's four major river deltas. Glob. Chang. Biol. 29 (8), 2286–2300.
Wu, M.-H., Chen, S.-Y., Chen, J.-W., Xue, K., Chen, S.-L., Wang, X.-M., Chen, T., Kang, S.-C., Rui, J.-P., Thies, J.E., Bardgett, R.D., Wang, Y.-F., 2021. Reduced microbial stability in the active layer is associated with carbon loss under alpine permafrost degradation. PNAS 118 (25).
Xu, L., He, N., Li, X., Cao, H., Li, C., Wang, R., Wang, C., Yao, M., Zhou, S., Wang, J., Soininen, J., 2021. Local community assembly processes shape β-diversity of soil phoD-harbouring communities in the Northern Hemisphere steppes. Glob. Ecol. Biogeogr. 30 (11), 2273–2285.
Ya, T., Du, S., Li, Z., Liu, S., Zhu, M., Liu, X., Jing, Z., Hai, R., Wang, X., 2021. Successional Dynamics of Molecular Ecological Network of Anammox Microbial Communities under Elevated Salinity. Water Res. 188, 116540.
Yang, P., Zhang, L., Lai, D.Y.F., Yang, H., Tan, L., Luo, L., Tong, C., Hong, Y., Zhu, W., Tang, K.W., 2022b. Landscape Change Affects Soil Organic Carbon Mineralization and Greenhouse Gas Production in Coastal Wetlands. Global Biogeochemical Cycles 36, e2022GB007469.
Yang, L., Ning, D., Yang, Y., He, N., Li, X., Cornell, C.R., Bates, C.T., Filimonenko, E., Kuzyakov, Y., Zhou, J., Yu, G., Tian, J., 2022a. Precipitation balances deterministic and stochastic processes of bacterial community assembly in grassland soils. Soil Biol. Biochem. 168, 108635.
Yu, Y., Shi, Y.u., Li, M., Wang, C., Zhang, L., Sun, Z., Lei, B., Miao, Y., Wang, W., Liu, B., Zheng, J., Chu, H., Han, S., 2022. Land-use type strongly affects soil microbial community assembly process and inter-kingdom co-occurrence pattern in a floodplain ecosystem. Appl. Soil Ecol. 179, 104574.
Yuan, M.M., Guo, X., Wu, L., Zhang, Y.a., Xiao, N., Ning, D., Shi, Z., Zhou, X., Wu, L., Yang, Y., Tiedje, J.M., Zhou, J., 2021. Climate warming enhances microbial network complexity and stability. Nat. Clim. Chang. 11 (4), 343–348.
Zhang, G., Bai, J., Tebbe, C.C., Zhao, Q., Jia, J., Wang, W., Wang, X., Yu, L.u., 2021a. Salinity controls soil microbial community structure and function in coastal estuarine wetlands. Environ. Microbiol. 23 (2), 1020–1037.
Zhang, Z., Lu, Y., Wei, G., Jiao, S., Zambrano, M.M., 2022. Rare Species-Driven Diversity-Ecosystem Multifunctionality Relationships are Promoted by Stochastic Community. Assembly. 13 (3), e0044922.
Zhang, P., Yang, Z., Wu, J., 2021b. Livestock grazing promotes ecosystem multifunctionality of a coastal salt marsh. J. Appl. Ecol. 58 (10), 2124–2134.
Zhang, Y., Zhang, L., Kang, Y., Li, Y., Chen, Z., Li, R., Tian, C., Wang, W., Wang, M., 2021c. Biotic homogenization increases with human intervention: implications for mangrove wetland restoration. Ecography 2022, e05835.
Zhao, S., Banerjee, S., White, J.F., Liu, J.-J., Zhou, N.a., Tian, C.-Y., 2022b. High salt stress increases archaeal abundance and network connectivity in saline agricultural soils. Catena 217, 106520.
Zhao, Z., Li, H., Sun, Y.i., Zhan, A., Lan, W., Woo, S.P., Shau-Hwai, A.T., Fan, J., 2022c. Bacteria versus fungi for predicting anthropogenic pollution in subtropical coastal sediments: Assembly process and environmental response. Ecol. Ind. 134, 108484.
Zhao, Q., Wang, Y., Xu, Z., Yun, J., Yu, Z., 2022a. Unravelling how biochar and dung amendments determine the functional structure and community assembly related to methane metabolisms in grassland soils. Biochar 4, 49.
Zhou, J., Ning, D., 2017. Stochastic Community Assembly: Does It Matter in Microbial Ecology? Microbiol. Mol. Biol. Rev. 81, Number 4.
Zhou, Y., Sun, B., Xie, B., Feng, K., Zhang, Z., Zhang, Z., Li, S., Du, X., Zhang, Q.i., Gu, S., Song, W., Wang, L., Xia, J., Han, G., Deng, Y.e., 2021. Warming reshaped the microbial hierarchical interactions. Global Chang Biology 27 (24), 6331–6347.
Zhou, Z., Wang, C., Luo, Y., 2020. Meta-analysis of the impacts of global change factors on soil microbial diversity and functionality. Nat. Commun. 11, 3072.
Zhu, X., Zhang, L., Zuo, Y., Liu, J., Yu, J., Yuan, F., Wang, N., He, L., Wang, Y., Guo, Z., Sun, Y., Song, Y., Song, C., Guo, D., Xu, X., 2021. Wetland reclamation homogenizes microbial properties along soil profiles. Geoderma 395, 115075.